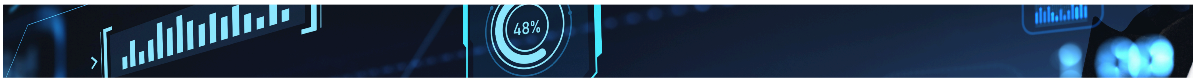
1.1 The Central Role of Time in Financial Analytics



A defining feature of financial analytics is that financial outcomes are realised **across time**, rather than at a single point.

Cash flows occur in different periods. Asset prices evolve continuously. Investment returns depend on both magnitude and timing. Risk is inherently dynamic, changing as market conditions and information evolve.

As a result, the timing of financial events affects valuation, risk assessment, and decision-making. Two investments with the same total payoff can have very different economic value depending on when those payoffs occur. This makes time an essential dimension of financial analysis, not a secondary consideration.

Time Series Data in Finance

Financial data is therefore most often observed and analysed as **time series data**.

Time series data consists of observations recorded sequentially over time, where the ordering of observations carries information. In finance, this includes:

- Daily or intraday asset prices
- Returns observed at regular intervals
- Interest rates, exchange rates, and yields over time
- Economic and financial indices measured monthly or quarterly

Time series data exhibits features such as trends, cycles, persistence, and volatility clustering. These characteristics mean that standard cross-sectional analysis is often insufficient, and specialised tools are required to analyse financial behaviour over time.